

Relative Performance Feedback and Redistribution Preferences: Evidence from an Online Experiment*

Isabella Valencia

Paris School of Economics

valenciaisabellap@gmail.com

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Abstract

This paper examines how information about relative performance affects distributive preferences and beliefs about the role of effort versus luck in generating inequality. Using an online experiment, participants complete a code-recognition task and are randomly assigned to one of three feedback conditions: no feedback, told they performed *above* a randomly matched peer, or told they performed *below* a randomly matched peer. They then act as impartial spectators and distribute a fixed sum between two workers whose inequality arose from either performance (merit scenario) or a lottery (luck scenario). Participants told they outperformed their peer redistribute approximately \$0.60 less to the lower-earning worker in the merit scenario compared to the no-feedback baseline, and score 0.8 points higher on an effort-attribution belief scale (0–10). These results are consistent with self-serving attribution: learning that one outperformed a peer leads to stronger beliefs that inequality reflects effort, which in turn reduces redistribution demand.

Keywords: redistribution, relative performance, self-serving attribution, fairness, online experiment

JEL codes: C91, D31, D63, D91

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1 Introduction

A long-standing question in political economy is why individuals differ in their preferences for redistribution. One prominent explanation emphasises the role of *beliefs about income determinants*: people who believe that inequality arises from effort and talent tend to favour less redistribution, while those who attribute it to luck or circumstance tend to favour more (Piketty, 1995; Alesina and Angeletos, 2005; Bénabou and Tirole, 2006).

A natural follow-up question is what shapes those beliefs. Social comparisons are ubiquitous: workers learn their rank in a wage distribution, students receive class rankings, and online platforms continuously display relative performance metrics. Does learning where one stands in a distribution affect how one interprets the sources of inequality — and through that channel, redistribution preferences?

This paper provides experimental evidence on this mechanism. Participants complete a performance task and are randomly assigned to one of three conditions: (i) *no feedback* about relative standing; (ii) told they performed *above* a randomly matched peer (*above* group); or (iii) told they performed *below* a matched peer (*below* group). Subsequently, all participants act as impartial spectators and decide how to distribute a fixed monetary sum between two workers, in two scenarios: one where inequality arose from differential performance (merit), and one where it arose from a lottery (luck).

The key findings are: (1) Participants told they outperformed a peer attribute significantly more of their own task outcome to effort ($\overline{EL}_{\text{above}} = 7.6$) compared to both the no-feedback ($\overline{EL}_{\text{nofb}} = 6.8$) and below ($\overline{EL}_{\text{below}} = 6.3$) groups on a 0–10 scale. (2) The same group redistributes less in the merit scenario (\$1.87 vs. \$2.47 and \$2.39 in the other groups). (3) The effect on redistribution is not statistically significant for the below group relative to the no-feedback baseline, suggesting the response is asymmetric.

These results are consistent with self-serving attribution (Hastorf and Cantril, 1954; Miller and Ross, 1975): individuals who learn they outperformed a peer update upward their beliefs about the role of effort, and this updated belief translates into lower redistribution in the merit scenario.

The rest of the paper is structured as follows. Section 2 reviews the related literature. Section 3 describes the experimental design. Section 4 presents the results. Section 5 concludes.

2 Related Literature

This paper contributes to three strands of literature.

Beliefs about inequality and redistribution. A foundational insight from Piketty (1995) is that beliefs about social mobility and the role of effort versus luck determine preferences for redistribution in equilibrium. Alesina and Angeletos (2005) provide empirical evidence: Americans attribute poverty more to individual failure than Europeans, which partly explains the cross-country gap in redistribution. Bénabou and Tirole (2006) model how self-serving motivated reasoning sustains two equilibria — a “belief in a just world” and a more luck-oriented view — with different steady-state redistribution levels. Our experiment directly tests the mechanism by which an informational shock (relative performance feedback) updates beliefs and behaviour.

Impartial spectator and fairness views. Cappelen et al. (2007) and Cappelen et al. (2013) use spectator designs to classify subjects into fairness types (libertarian, meritocratic, egalitarian) and show that a majority of individuals accept some redistribution away from luck-generated inequality even when indifferent to merit-generated inequality. We adapt their two-scenario design (merit and luck) to test whether third-party redistribution decisions are affected by information about the spectator’s own relative performance — a test of whether the spectator role is truly impartial.

Self-serving attribution and social comparisons. Self-serving attribution refers to the tendency to attribute successes internally (effort) and failures externally (luck or circumstance) (Miller and Ross, 1975; Mezulis et al., 2004). In an economic context, Babcock and Loewenstein (1995) document that parties in bargaining inflate self-assessments of a fair outcome in a self-serving direction. Closer to our design, Deffains et al. (2016) show that individuals who experience unfair treatment become more supportive of redistribution. We add to this literature by focusing on the *comparative* dimension: the same absolute performance level can produce different attribution depending on whether one learns one stands above or below a peer.

3 Experimental Design

The experiment was conducted online using JATOS and Prolific. Participants completed two parts: a performance task (Part 1) and a distribution task (Part 2). The study took approximately 7–10 minutes and participants were compensated according to standard Prolific rates plus a bonus determined by a randomly selected distribution decision.

3.1 Part 1: Code Recognition Task

Participants completed a code-recognition task over five rounds of 45 seconds each. In each round, a target code was displayed at the top of the screen. Participants were shown a grid of codes and had to check as many instances of the target as possible. Scoring: +1 point per correct identification, -1 point per false alarm, 0 points for a miss. The maximum possible score was 200 points.

Before seeing their results, participants answered three pre-feedback self-report questions: (i) effort exerted (slider, 0–10); (ii) guessed score (numeric, range -200 to 200); (iii) relative performance prediction (“I would perform better or the same” vs. “I would perform worse than the other participant”).

3.2 Treatments: Relative Performance Feedback

After the self-report, participants were randomly assigned to one of three feedback conditions.

1. **No feedback.** Participants were simply informed they could proceed to Part 2.
2. **Above.** Participants were shown their score and told: “You performed above your randomly matched peer.”
3. **Below.** Participants were shown their score and told: “You performed below your randomly matched peer.”

Matching was done using previously collected scores from a pilot wave, ensuring that the “above” vs. “below” assignment was credible given each participant’s actual score. Approximately half of the relative-feedback group was told they outperformed (above) and half that they underperformed (below).

3.3 Part 2: Distribution Decisions

After the feedback screen, all participants acted as impartial spectators and made two sequential redistribution decisions.

Decision 1 — Merit Scenario

Participants read:

“Two previous participants, Worker A and Worker B, completed the same task you just completed. [*Worker A earned \$8; Worker B earned \$0 because they*

performed worse/. You must decide how to distribute a total of \$8 between them.”

The distribution was presented as 9 discrete options ranging from \$0 to Worker B (keeping full inequality) to \$8 to Worker B (reversing inequality), with \$4 each as the equal split. The outcome variable is the amount allocated to the lower earner, Worker B (`redist_merit`, range \$0–\$8).

Decision 2 — Luck Scenario

After completing a demographics survey (including the effort–luck belief slider), participants saw a second decision:

“Worker C and Worker D completed the same task. This time, earnings were assigned by a **lottery** with no regard for performance — Worker C won and received \$8; Worker D received \$0. How would you reallocate the total of \$8?”

The outcome variable is the amount allocated to Worker D (`redist_luck`).

3.4 Key Outcome and Mediator Variables

- **Redistribution (merit):** dollars allocated to Worker B (\$0–\$8).
- **Redistribution (luck):** dollars allocated to Worker D (\$0–\$8).
- **Effort–luck belief (`effort_luck`):** a slider recorded after the merit decision: “To what extent do you think the initial income of participants in the previous task was determined by effort versus luck?” (0 = 100% luck, 10 = 100% effort). This variable serves as the hypothesised mediator.

3.5 Sample

The analysis sample consists of $N = 134$ participants after excluding those with zero score and zero clicks (non-engagers) and those with implausibly high scores (>\$5 allocated to Worker B). Table 1 presents descriptive statistics by treatment group.

4 Results

4.1 Effort–Luck Beliefs

Figure 1 (left panel) shows mean effort–luck belief by treatment group. Participants told they outperformed a peer score markedly higher on the effort-attribution scale ($\overline{EL} = 7.6$) relative to the no-feedback group ($\overline{EL} = 6.8$) and the below group ($\overline{EL} = 6.3$).

Table 2 (columns 1–2) presents OLS regressions of `effort_luck` on treatment dummies (baseline: no feedback), with and without demographic controls. Being told one outperformed a peer increases the effort-attribution score by approximately [XX] points (SE = [XX], $p = [XX]$), while being told one underperformed decreases it by approximately [XX] points, though this coefficient is [not] statistically significant. The pattern is consistent with self-serving attribution: winners attribute the outcome to effort; losers do not systematically update in the opposite direction.

4.2 Redistribution in the Merit Scenario

The right panel of Figure 1 shows mean redistribution in the merit scenario. The above group redistributes the least (\$1.87), compared to the no-feedback baseline (\$2.47) and the below group (\$2.39).

Columns 3–4 of Table 2 report OLS regressions of `redist_merit` on treatment dummies. Being told one outperformed reduces redistribution by approximately [XX] cents (SE = [XX], $p = [XX]$) relative to the no-feedback group, corresponding to a [XX]% reduction from the no-feedback mean. The below coefficient is small and statistically indistinguishable from zero, confirming the asymmetry visible in Figure 1. Results are robust to controlling for age, gender, and task score (column 4).

4.3 Heterogeneity: Surprise

To probe the mechanism further, we classify participants in the relative group by whether the feedback was surprising. *Positively surprised* participants predicted they would perform worse than or the same as their peer but were told they outperformed (`above_positively_surprised` = 1). *Negatively surprised* participants predicted they would outperform but were told they underperformed (`below_negatively_surprised` = 1).

If the self-serving attribution channel is operative, the redistribution effect should be driven primarily by those who were unexpectedly told they outperformed, as these individuals face the largest positive surprise. Column 5 of Table 2 reports this regression. [Results to be discussed once final sample is available.]

4.4 Redistribution in the Luck Scenario

As a placebo and secondary outcome, we examine redistribution in the luck scenario where earnings are explicitly attributed to a lottery. If feedback affects redistribution through effort-attribution beliefs, its effect should be attenuated (or absent) in the luck scenario, where performance is irrelevant by design. [Results to be reported.]

5 Conclusion

This paper provides experimental evidence that relative performance feedback shapes both beliefs about the role of effort in generating inequality and downstream redistribution preferences. Participants told they outperformed a peer attribute more of the task outcome to effort and redistribute less in a merit scenario, consistent with self-serving attribution updating meritocratic beliefs. The effect is asymmetric: underperformers do not significantly increase redistribution relative to a no-feedback baseline.

These findings have implications for the study of inequality and redistribution in a world where performance comparisons are increasingly salient — on professional platforms, social networks, and in educational settings. The self-serving updating documented here suggests that rising relative visibility of performance rankings could systematically reduce support for redistribution among those who “win” such comparisons, potentially contributing to the polarisation of redistribution preferences.

Future work should (i) increase sample size to improve precision, particularly for the heterogeneity analysis; (ii) test whether the effect persists when participants are given their absolute score alongside relative feedback; and (iii) examine whether the mediation through effort–luck beliefs is statistically confirmed with a formal mediation analysis.

Tables and Figures

Table 1: Descriptive Statistics by Treatment Group

	No Feedback (1)	Above (2)	Below (3)	p -value (4)	
Redistribution — merit (\$)	2.47	1.87	2.39	[XX]	<i>Notes.</i> Column (4)
Redistribution — luck (\$)	[XX]	[XX]	[XX]	[XX]	
Effort–luck belief (0–10)	6.8	7.6	6.3	[XX]	
Score (task points)	[XX]	[XX]	[XX]	[XX]	
Age	[XX]	[XX]	[XX]	[XX]	
Female (%)	[XX]	[XX]	[XX]	[XX]	
Predicted better (%)	[XX]	[XX]	[XX]	[XX]	
N	[XX]	[XX]	[XX]	134	

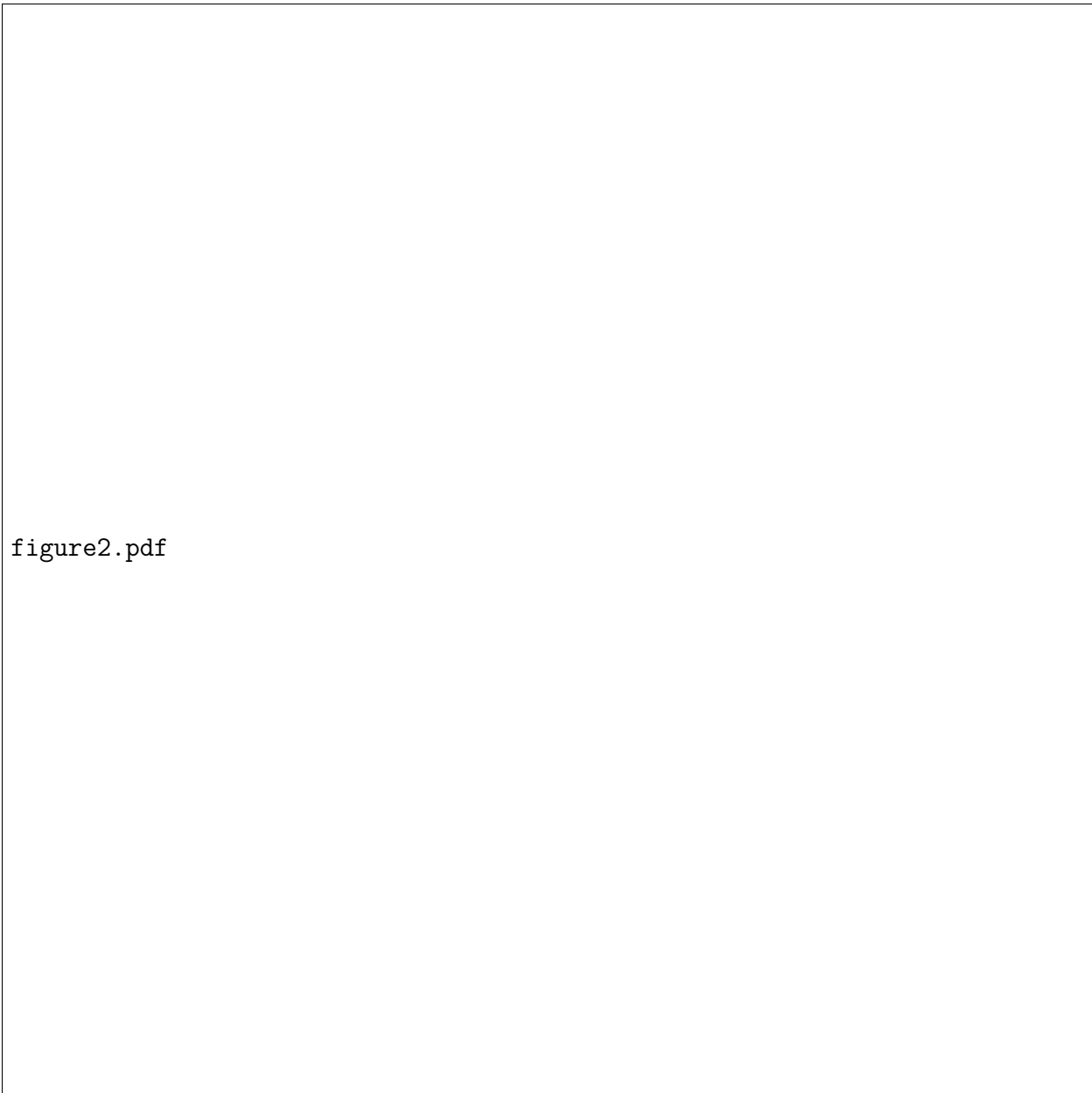
reports the p -value from a one-way ANOVA (continuous) or chi-squared test (binary). Redistribution values are in USD.

Table 2: OLS Regressions: Effort–Luck Belief and Merit Redistribution

	Outcome: effort–luck belief		Outcome: redistribution — merit (\$)		
	(1)	(2)	(3)	(4)	(5)
Told: Above	[XX]* ([XX])	[XX]* ([XX])	[XX]* ([XX])	[XX]* ([XX])	—
Told: Below	[XX] ([XX])	[XX] ([XX])	[XX] ([XX])	[XX] ([XX])	—
Positively surprised (above $\times$ $\neg$ pred)	—	—	—	—	[XX] ([XX])
Negatively surprised (below $\times$ pred)	—	—	—	—	[XX] ([XX])
Age	No	Yes	No	Yes	No
Female	No	Yes	No	Yes	No
Score	No	Yes	No	Yes	No
N	[XX]	[XX]	[XX]	[XX]	[XX]
R^2	[XX]	[XX]	[XX]	[XX]	[XX]

Notes. Heteroskedasticity-robust standard errors (HC3) in parentheses. Baseline category: no feedback. * $p < 0.05$. Column (5) restricts to the relative-feedback group and replaces treatment dummies with the surprise indicators.

Figure 1: Mean Effort Attribution and Merit Redistribution by Treatment



Notes. Bar heights correspond to group means. Whiskers represent ± 1 standard error of the mean. Left panel: effort–luck belief slider (0 = 100% luck; 10 = 100% effort). Right panel: amount allocated to the lower-earning worker in the merit scenario (USD, range \$0–\$5). $N = 134$.

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